CP and then sent through an additive white Gaussian noise (AWGN) channel. At the receiver, the CP is removed, serial form of signal is converted into parallel form and FFT is applied to each symbol for analysing the signal in the frequency field. After demodulation, it is cross-correlated with a local oscillator signal, which is dislodged in time. Finally, the picked-up signal is compared with the threshold for sensing of the spectrum. A hypothesis is formulated that if the received signal is larger than the threshold, there will be detection, otherwise not.

The results are thoroughly analysed by using a mathematical model of cognitive OFDM system and code is implemented using MATLAB programs (filter_tx.m, filter_tx_rx.m, pd_pfa.m, pd_pfa_cp.m, pd_pfa_rx.m, pd_pfa_tx.m, pd_snr.m, pd_snr_cp.m) included in this month's EFY DVD.

Analysis results

BER vs SNR for OFDM with and without CP. Analysis of BER vs SNR for OFDM with and without CP is shown in Fig. 2. BER decreases as SNR increases. From 0dB to 3dB, BER for OFDM with and without CP is approximately equal. But as SNR increases beyond 3dB, BER performance of OFDM with CP is a lot better than OFDM without CP. Analytical results prove that the OFDM system with CP is much more efficient and authentic with less ISI and BER than OFDM without CP.

BER vs SNR for OFDM with filter at transmitter and receiver. Analysis of BER vs SNR for OFDM with a filter at both the transmitter and the receiver is shown in Fig. 3. For OFDM system with filter at the transmitter, BER decreases with the increase of SNR. But OFDM with filter at both transmitter and receiver sides gives a constant BER for all SNRs. This proves that OFDM with filter at the transmitter is more beneficial than OFDM with filter at both

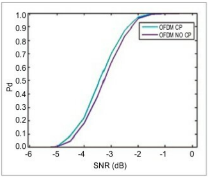

Fig. 4: Pd vs SNR for OFDM with and without CP

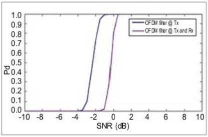

Fig. 5: Pd vs SNR for OFDM with filter at transmitter and receiver

the transmitter and the reciever.

Pd vs SNR for OFDM with and without CP. Analysis of Pd vs SNR for OFDM with and without CP is shown in Fig. 4. Pd increases with the increase of SNR. OFDM system with CP is capable of detecion for smaller SNRs, whereas detection for OFDM without CP needs a high SNR.

Pd vs SNR for OFDM with filter at transmitter and receiver. Pd vs SNR plots for an OFDM system with a filter at the transmitter, and an OFDM system with a filter at both the transmitter and the receiver, are shown in Fig. 5. Pd increases with the increase of SNR. OFDM system with a filter at the transmitter is capable of detection for smaller SNRs, whereas OFDM system with a filter at both the transmitter and the receiver needs a high SNR for detection.

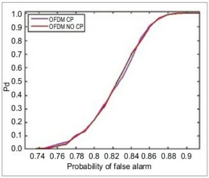

Fig. 6: Pd vs Pfa for OFDM with and without CP

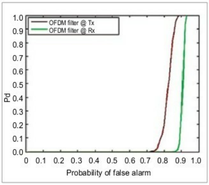

Fig. 7: Pd vs Pfa for OFDM with filter at transmitter and receiver

Pd vs Pfa for OFDM with and without CP. Analysis of Pd vs Pfa for OFDM with and without CP is shown in Fig. 6. Pd increases with the increase of Pfa. For OFDM with CP the probability of missed detection is higher, even for lower Pfa, whereas for OFDM without CP the Pfa requirement is more for missed detection, which means the effect of Pfa in OFDM without CP is less than in OFDM with CP.

Pd vs Pfa for OFDM with filter at transmitter and receiver. Analysis of Pd vs Pfa for OFDM with filter at transmitter and receiver both is shown in Fig. 7. OFDM with filter at transmitter shows a better performance than OFDM with filter at both transmitter and receiver. EFY

EFY Note

The MATLAB programs of this project are included in this month's EFY DVD and are also available for free download at *source.efymag.com*


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